Icarus rev 0.2 — dual-channel triple-shutdown steering torque-overlay unit

CH-A (F280049C-Q1) computes the overlay; CH-B (F280049C-Q1) monitors it; CH-C is an analog guard with no software.
Overlay reaches the EPS only through relays K1 AND K2; each coil needs CH-A high side + CH-B low side + CH-C guard FET.
With no power or any enable missing, NC contacts connect the OEM torque sensor straight to the EPS.
Safety concept: ../docs/04_TECHNICAL_SAFETY_CONCEPT.md. Values marked TBD-DATA wait for logger measurements.

power

Power: input protection and two independent supply chains

File: power.kicad_sch

ch_a

CH-A: F280049C-Q1 control (computes and drives the overlay)

File: ch_a.kicad_sch

ch_b

CH-B: F280049C-Q1 monitor (checks; owns the low-side coil enable)

File: ch_b.kicad_sch

overlay

Overlay path (CH-A analog, +5V0_A)

File: overlay.kicad_sch

guard

CH-C analog guard (no software, +5V0_B)

File: guard.kicad_sch

relays

Signal relays K1/K2 (cascaded) and cross-driven coils

File: relays.kicad_sch

io

Vehicle connector, CAN bus, inputs and monitors

File: io.kicad_sch

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Dual-channel steering torque-overlay unit, Honda Insight EPS

Sheet: /
File: icarus.kicad_sch

Title: Icarus — top level

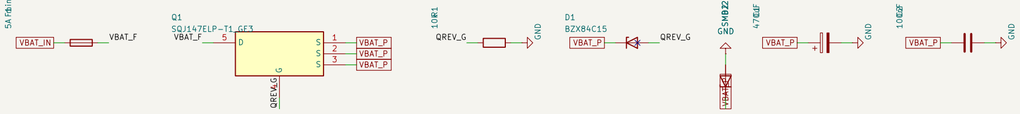
Size: A3	Date: 2026-11-01	**Rev: 0.2**
KiCad E.D.A. 10.0.0		Id: 1/8

Power: input protection and two independent supply chains

Everything after VBAT_P is duplicated per channel (SM-16): buck -> LDO -> supervisors.

Input protection (shared)

5A fast-blow fuse



Supply chain A: LMR38020-Q1 5.0 V buck, TPS7B81-Q1 3.3 V LDO, TPS3702-Q1 5 V window



Supply chain B: LMR38020-Q1 5.0 V buck, TPS7B81-Q1 3.3 V LDO, TPS3702-Q1 5 V window

Power flags (rails driven by regulators / the protection FET)

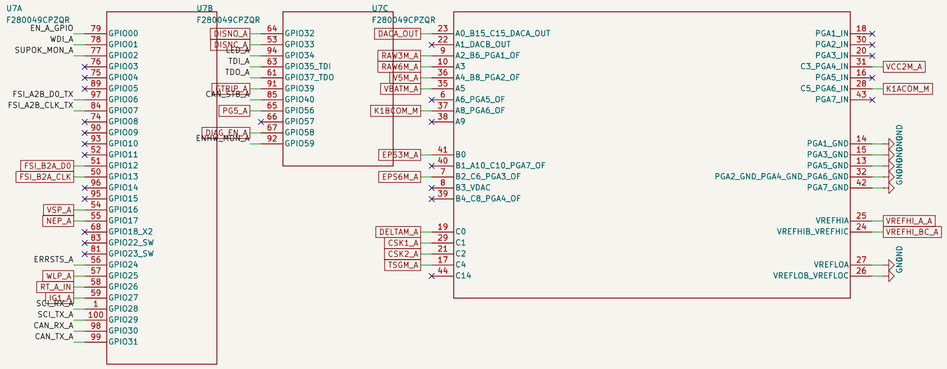
Generated by hardware/gen/make_sch.py from gen/design.py - do not hand-edit
Dual-channel steering torque-overlay unit, Honda Insight EPS

Sheet: /power/		
File: power.kicad_sch		
Title: Icarus — Power: input protection and two independent supply cha		
Size: A2	Date: 2026-11-01	**Rev: 0.2**
KiCad E.D.A. 10.0.0		Id: 2/8

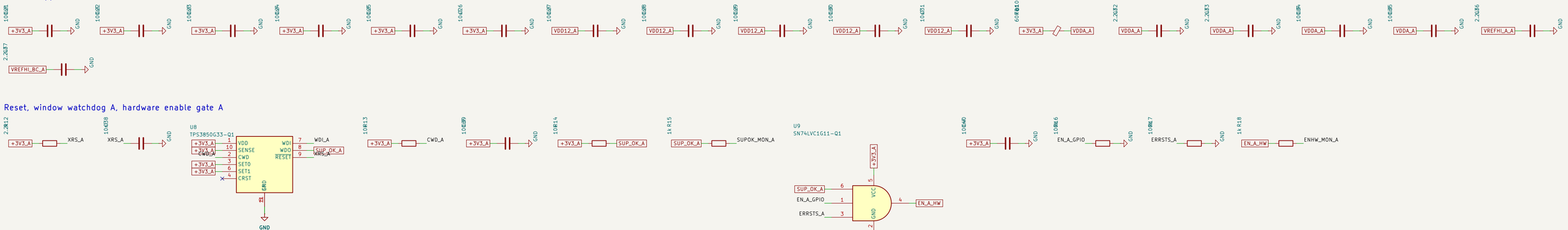
CH-A: F280049C-Q1 control (computes and drives the overlay)

Pin mux from SPRS945 Table 5-6. Window watchdog TPS3850-Q1: CWD 10k to VDD, SET0 = SET1 = 1 -> 1.85 ms open / 11 ms timeout, service every ~5 ms.

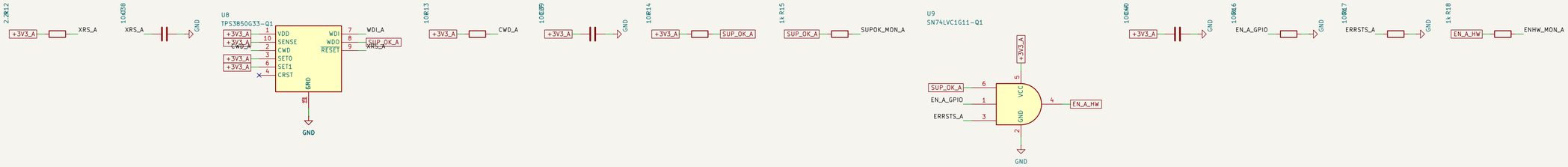
MCU A



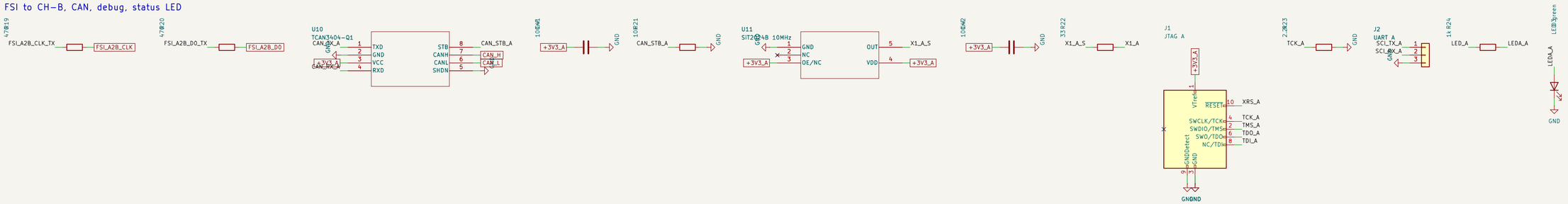
MCU A supplies and reference



Reset, window watchdog A, hardware enable gate A



FSI to CH-B, CAN, debug, status LED



Generated by hardware/gen/make_sch.py from gen/design.py - do not hand-edit
Dual-channel steering torque-overlay unit, Honda Insight EPS

Sheet: /ch_a/
File: ch_a.kicad_sch

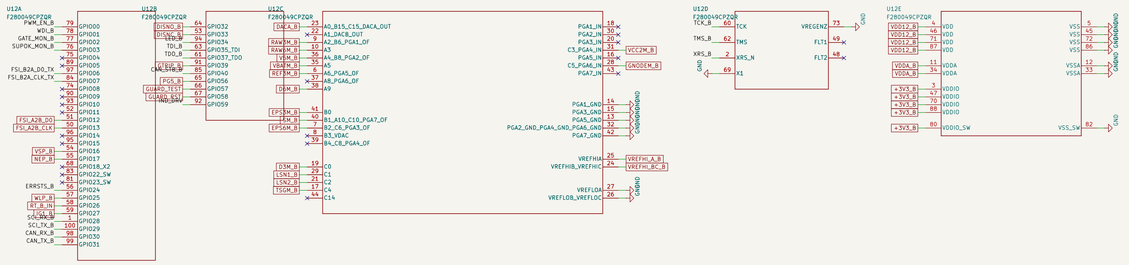
Title: Icarus - CH-A: F280049C-Q1 control (computes and drives the o

Size: A2	Date: 2026-11-01	Rev: 0.2
KiCad E.D.A. 10.0.0		Id: 3/8

CH-B: F280049C-Q1 monitor (checks; owns the low-side coil enable)

Pin mux from SPRS945 Table 5-6. Window watchdog TPS3850-Q1: CWD 10k to VDD, SET0 = SET1 = 1 -> 1.85 ms open / 11 ms timeout, service every ~5 ms.

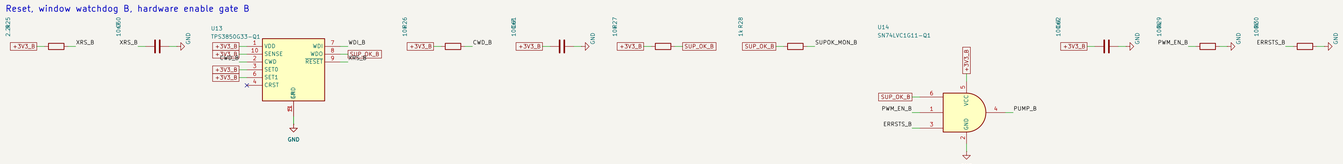
MCU B



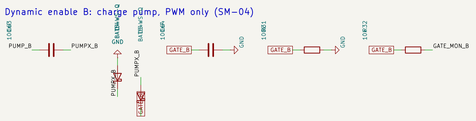
MCU B supplies and reference



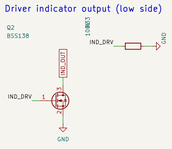
Reset, window watchdog B, hardware enable gate B



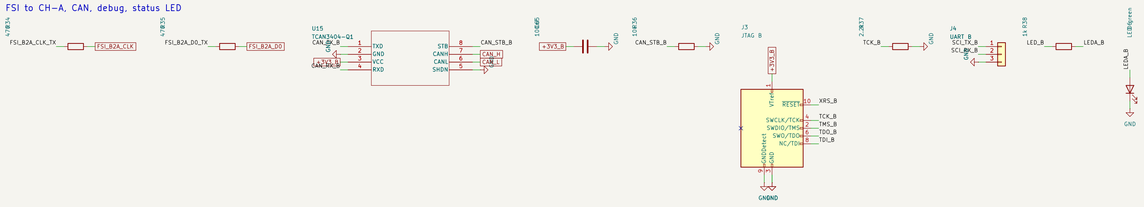
Dynamic enable B: charge pump, PWM only (SM-04)



Driver indicator output (low side)



FSI to CH-A, CAN, debug, status LED



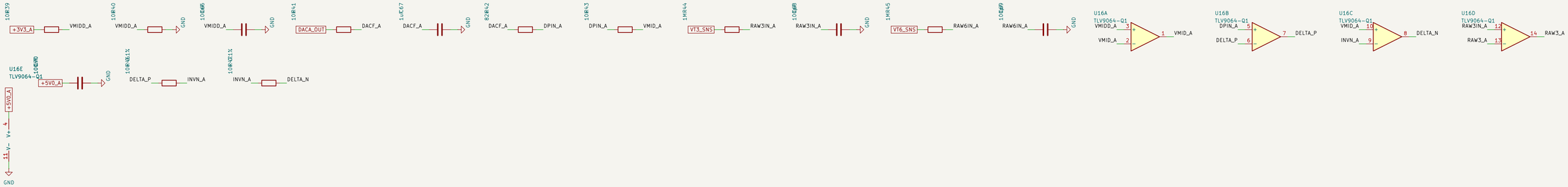
Generated by hardware/gen/make_sch.py from gen/design.py — do not hand-edit
Dual-channel steering torque-overlay unit, Honda Insight EPS

Sheet: /ch_b/	
File: ch_b.kicad_sch	
Title: Icarus — CH-B: F280049C-Q1 monitor (checks; owns the low-side coil enable)	
Size: A1	Date: 2026-11-01
KiCad E.D.A. 10.0.0	**Rev: 0.2** Id: 4/8

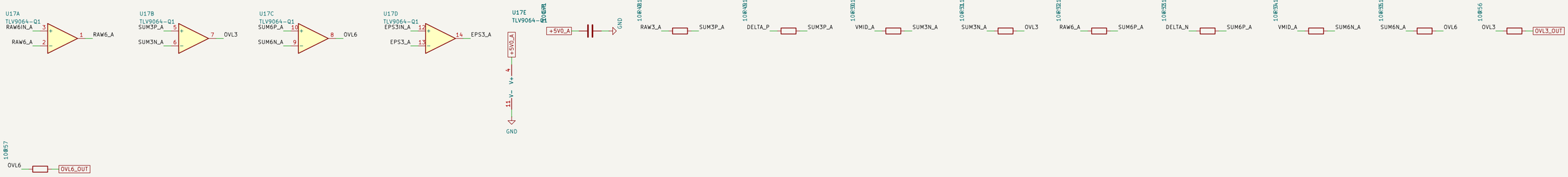
Overlay path (CH-A analog, +5V0_A)

EPS gets OVL = RAW + (DELTA − VMID), DELTA = VMID + k(DAC − VMID), k = 10k/92k = 0.109 -> Delta_max = +/−0.18 V (TBD-DATA, SM-05). One DAC + inverter makes VT6 = −Delta (SM-06). Summing resistors 0.1 %.

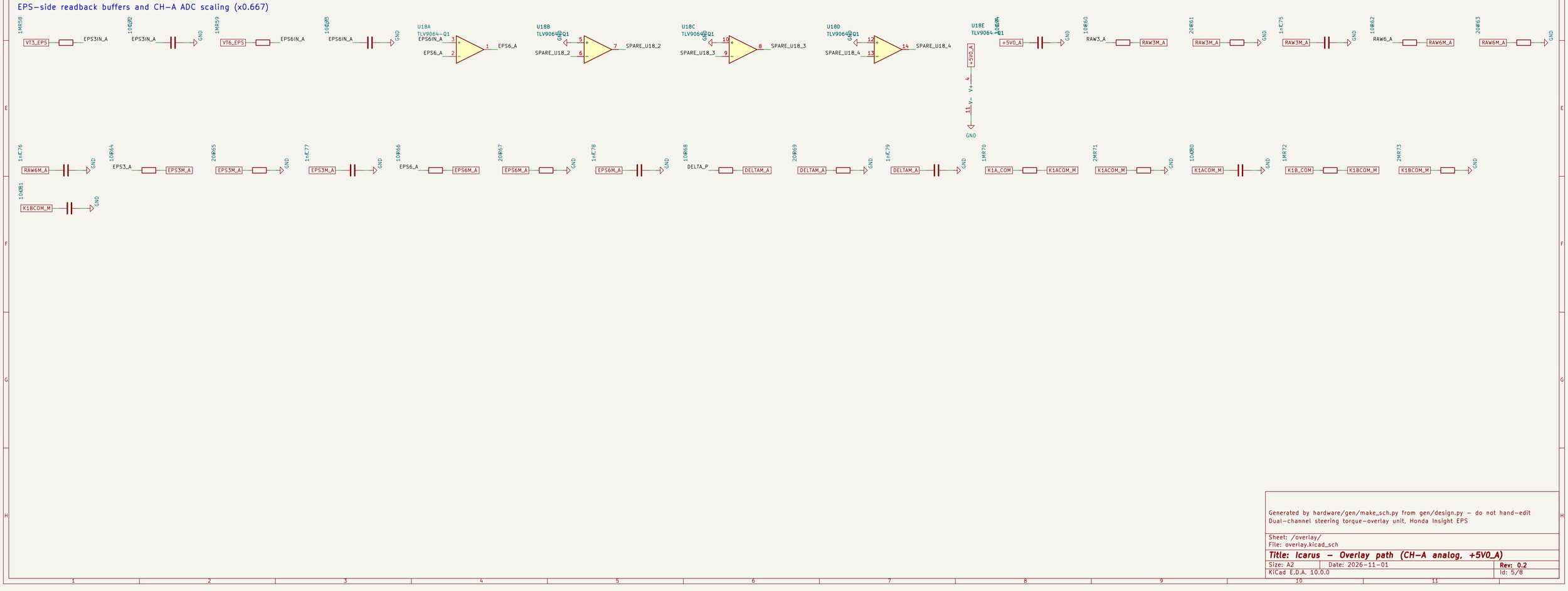
References and Delta generation



Summers: OVL3 = RAW3 + DELTA_P − VMID, OVL6 = RAW6 + DELTA_N − VMID



EPS-side readback buffers and CH-A ADC scaling (x0.667)



Generated by hardware/gen/make_sch.py from gen/design.py — do not hand-edit
Dual-channel steering torque-overlay unit, Honda Insight EPS

Sheet: /overlay/
File: overlay.kicad_sch

Title: Icarus — Overlay path (CH-A analog, +5V0_A)

Size: A2 | Date: 2026-11-01 | **Rev: 0.2**
KiCad E.D.A. 10.0.0 | Id: 5/8

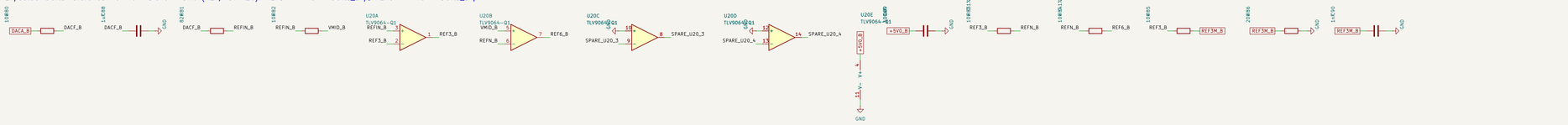
CH-C analog guard (no software, +5V0_B)

Trips when the overlay actually applied (VT_EPS − VT_SNS) differs from CH-B's own expected Delta by > 0.25 V on either line (R2 / SM-20), or the VT3+VT6 sum moves > 0.10 V (TBD-DATA). A trip latches until key-off and cuts every coil (SM-15). GUARD_TEST forces a trip for the key-on self-test (SM-17). VMID_B and the ladder are ratiometric to +3V3_B, the CH-B DAC reference.

Buffers (1 MOhm from the car side) and VMID_B



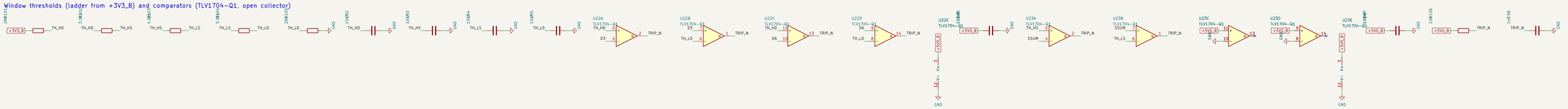
Expected-Delta reference from CH-B's own DAC (R2 / SM-20): REF3 = VMID − Delta_exp, REF6 = VMID + Delta_exp



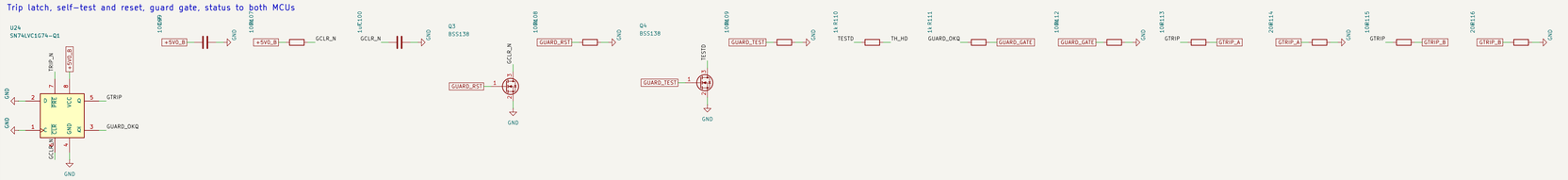
Difference amps: D3 = EPS3 − RAW3 + REF3 (= VMID when the applied Delta matches the expected), D6 likewise, S = (EPS3+EPS6) − (RAW3+RAW6) + VMID



Window thresholds (ladder from +3V3_B) and comparators (TLV1704-Q1, open collector)



Trip latch, self-test and reset, guard gate, status to both MCUs



CH-B ADC scaling (x0.667)



Generated by hardware/gen/make_sch.py from gen/design.py — do not hand-edit
Dual-channel steering torque-overlay unit, Honda Insight EPS

Sheet: /guard/
File: guard.kicad_sch

Title: Icarus — CH-C analog guard (no software, +5V0_B)

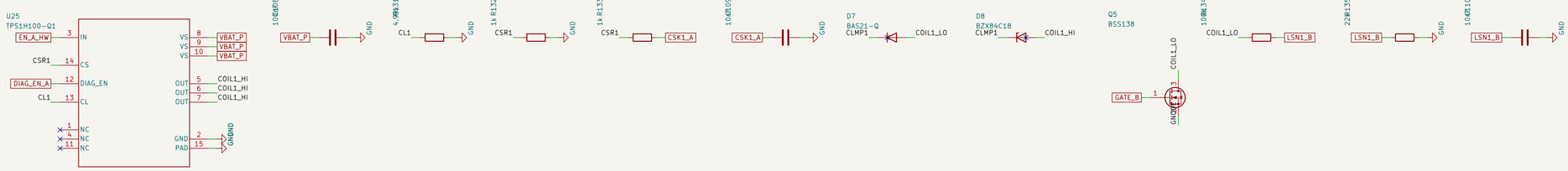
Size: A1 | Date: 2026-11-01 | **Rev: 0.2**

KiCad E.D.A. 10.0.0 | Id: 6/8

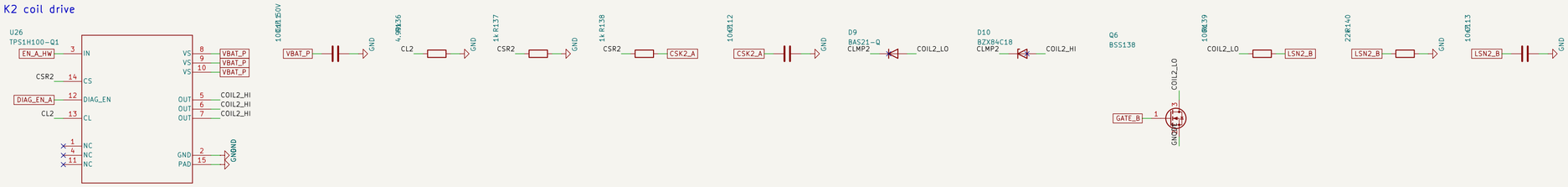
Signal relays K1/K2 (cascaded) and cross-driven coils

EPS sees the overlay only if K1 AND K2 are energized (SM-14). Each coil: high side from CH-A (TPS1H100-Q1), low side from CH-B (charge-pumped gate) and the CH-C guard FET in series.

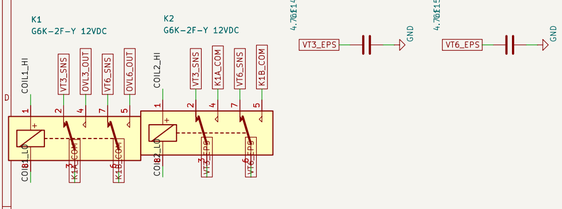
K1 coil drive



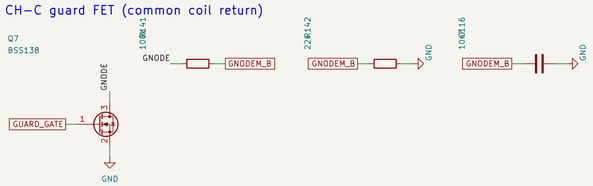
K2 coil drive



Relays: K1 picks overlay vs sensor, K2 picks K1 vs sensor



CH-C guard FET (common coil return)



Generated by hardware/gen/make_sch.py from gen/design.py — do not hand-edit
Dual-channel steering torque-overlay unit, Honda Insight EPS

Sheet: /relays/
File: relays.kicad_sch

Title: Icarus — Signal relays K1/K2 (cascaded) and cross-driven coils

Size: A2 | Date: 2026-11-01 | **Rev: 0.2**
KiCad E.D.A. 10.0.0 | Id: 7/8

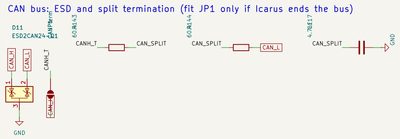
Vehicle connector, CAN bus, inputs and monitors

Every car-facing input has its own divider per channel (no shared resistor between CH-A and CH-B). Torque-sensor and Vcc2 taps are 1 MOhm (FSR-14).

Vehicle connector (Y-harness at EPS connector B; part TBD, TE MCP family as throttle J2)



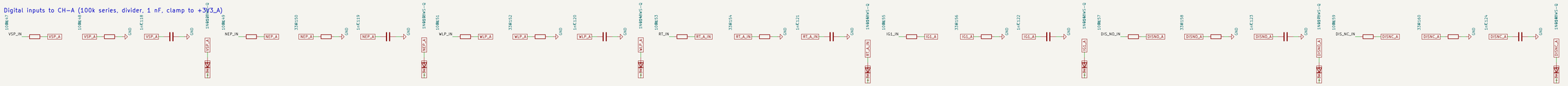
CAN bus: ESD and split termination (fit JP1 only if Icarus ends the bus)



Disengage switch wetting (NO + NC contacts to GND)



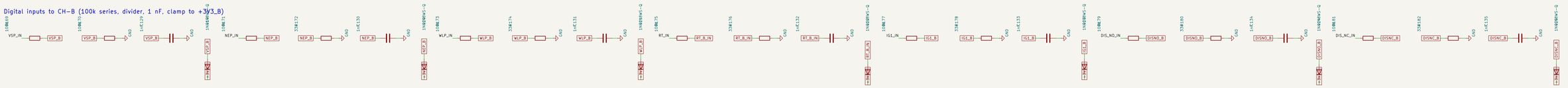
Digital inputs to CH-A (100k series, divider, 1 nF, clamp to +3V3_A)



Analog monitors to CH-A



Digital inputs to CH-B (100k series, divider, 1 nF, clamp to +3V3_B)



Analog monitors to CH-B



Generated by hardware/gen/make_sch.py from gen/design.py — do not hand-edit
Dual-channel steering torque-overlay unit, Honda Insight EPS

Sheet: /io/	
File: io.kicad_sch	
Title: Icarus — Vehicle connector, CAN bus, inputs and monitors	
Size: A1	Date: 2026-11-01
Rev: 0.2	
KiCad E.D.A. 10.0.0	Id: 8/8